OpenBuild — Executive Findings

Analytics engineering portfolio · 108K orders · 87K users · 2019–2022 · Medallion architecture · 28 tested transformations

1. Retention is structurally flat, not strategically broken

RESULT

~1% month-1 retention across all 48 cohorts (Jan 2019 – Dec 2022). Acquisition strategy did not move repurchase behavior in any cohort.

IMPLICATION

Marketing initiatives aimed at retention will not move the needle. The lever is product mix.

CAUSES

Cause	Weight	Status
Durable goods naturally have low repurchase frequency	High	Tested
Catalog skews toward one-time purchases vs. replenishables	High	Tested
Limited cross-sell mechanics in checkout	Medium	Hypothesis

Source models: mart_cohort_retention, dim_users, fct_orders · See Appendix A1

2. Laptops drive 2–3× the company refund rate

RESULT

Refund baseline: 4.97%. Top offenders are all laptops: ThinkPad × CA (21.3%), ThinkPad × GB (16.8%), MacBook Air × CA (16.3%). MacBook Air × US is the largest dollar leak: $365K refunded.

IMPLICATION

Refund-reduction initiatives should target laptops first. Geographic variation suggests fulfillment or returns-policy investigation by country.

CAUSES

Cause	Weight	Status
High-AOV products carry higher absolute return risk	High	Tested
Spec / expectation mismatch in laptops vs. accessories	High	Partially tested
Country-level returns policy or fulfillment variance	Medium	Partially tested
Localization issues (pricing, positioning)	Medium	Hypothesis

Source models: mart_refund_metrics, fct_orders, dim_country · See Appendix A2

3. Mobile is structurally a low-AOV channel

RESULT

Website = 96.8% of lifetime revenue. Mobile generates 17% of orders but only 3% of revenue. Mobile share moved from 2.95% (2019) to 3.93% (2022) — statistically real, business-trivially small.

IMPLICATION

Mobile investment thesis must target AOV, not order volume. Doubling mobile order volume only adds ~3 percentage points to total revenue at current AOV.

CAUSES

Cause	Weight	Status
Mobile users skew toward accessory purchases	High	Tested
Catalog product mix is desktop-shaped (laptops, monitors)	High	Tested
Mobile UX friction at high-value checkout	Medium	Hypothesis
Desktop preference for large / considered purchases	Low–Medium	Hypothesis

Source models: mart_channel_revenue, fct_orders, dim_platform · See Appendix A3

Full analysis, SQL, and tests: github.com/joyce92200/ecommerce-analytics-engineering

Appendix — Methodology

Each finding's metric definition, SQL logic, and caveats. All numbers reproducible from the repository above.

A1 — Cohort retention

Metric. Period-active retention: % of a cohort placing at least one order during month N after acquisition. Not "made a 2nd purchase ever" (different metric).

Cohort definition. cohort_month = DATE_TRUNC('month', MIN(purchase_ts)) per user.

```
retention_pct = active_users / cohort_size × 100
active_users = COUNT(DISTINCT user_id) per (cohort_month, months_since_acquisition)
cohort_size = active_users at months_since_acquisition = 0
```

Caveats. Cohorts after Dec 2021 are right-censored — month-12 retention is unobservable given the data window ending Dec 2022. Matrix reported only for fully observable cells. 3 orders (0.003%) with unparseable purchase timestamps excluded.

A2 — Refund metrics

Metric. Two metrics at (product × country) grain: *refund_rate_pct* = refunded orders / total × 100; *refunded_revenue_usd* = SUM(usd_price WHERE is_refunded = 1).

Refund definition. An order is refunded if refund_ts is non-null. Partial vs. full refunds are not distinguished in the source.

```
WHERE orders >= 100
ORDER BY refund_rate_pct DESC
LIMIT 10
```

Caveats. 100-order minimum is a statistical-reliability threshold. 4 orders with NULL country_code retained as a distinct segment so unknown-geo leakage stays visible. 18 orders with non-standard codes (EU, AP) mapped to Unclassified for referential integrity.

A3 — Channel revenue

Metric. Net revenue (gross minus refunded) by (purchase_month × purchase_platform) with within-month share %.

```
net_revenue_usd = SUM(usd_price WHERE is_refunded = 0)
share_of_month_pct = net_revenue / SUM(net_revenue) OVER (PARTITION BY purchase_month) × 100
```

Trend comparison. 2019 vs. 2022 share = aggregate all months in each year, then take the platform's share within that year.

Caveats. Channel attribution is platform-only (website vs. mobile app). Marketing channel attribution is unavailable (upstream column >99% null). AOV comparisons use net_revenue / orders across platforms.

How to verify

Clone the repository, install requirements, open notebooks/01_cohort_exploration.ipynb, and run all cells top-to-bottom. Each finding corresponds to a specific cell whose output matches the numbers reported. The 28 data quality tests in tests/test_models.sql validate every input model.